ReSolve: Astra-Guided CAD Patch Repair with Geometric Verification and Graph Search

Anonymous CVPR submission

Paper ID Draft

Abstract

Instruction-guided CAD editing requires changing a requested feature while preserving unrelated geometry. We study a frozen Astra model that produces original-coordinate program patches, combined with experience retrieval, graph search, execution feedback and patch repair. Candidates are checked for executable geometry, instruction-derived values, rendered-view semantic consistency, conditional finite-element response and robustness. On a locally defined 124-task BenchCAD holdout, the single Astra answer reaches 89 strict reference matches. The recorded full graph pipeline reaches 85, with 2 recoveries and 6 regressions; it does not improve agreement. Retrieval alone reaches 94, but this configuration was identified after observing results and its paired gain is nonsignificant. Of 35 incorrect initial answers, 24 pass the geometry, numeric and FEM checks, exposing a gap between target-free feasibility and instruction correctness. The study reports a reproducible patch-repair workflow and its observed limitations. No model is trained in this pipeline, and physical checks use explicit diagnostic scenarios rather than inferred operating conditions.

1. Introduction

A request to enlarge a bore, remove a bracket flange or add a blind shaft slot changes one part of a CAD construction while preserving the remaining shape. Figure 1 illustrates released source/target pairs. Program execution alone does not determine whether the requested edit was made: a valid solid may have the wrong feature dimensions or locations.

We implement ReSolve as a patch-editing and repair workflow around a frozen Astra model. The model receives source CAD, an instruction and permitted observations; it emits a typed patch against the original source. Execution produces the edited solid. When a check fails, a diagnosis conditions a replacement patch, which is again applied to the original source and re-evaluated. Search can explore alternative interpretations, backtrack and retain multiple candidates.

The central empirical question is whether target-free verification provides enough information to select better patches. We examine experience retrieval, diverse hypotheses, MCTS, Pareto selection, rendered-view judgement and conditional physical checks. In the recorded BenchCAD experiment, these components do not improve the initial 89/124 strict matches: the full configuration reaches 85/124. Many incorrect answers satisfy the available checks, while some correct answers fail them.

Contributions.

- An executable Astra-guided CAD patch-repair pipeline with original-source indexing, structured failure diagnoses and bounded graph exploration.
- A staged evaluation of retrieval, search and verification on a fixed local BenchCAD holdout, reporting recoveries, regressions, call counts and costs.
- An explicit separation between geometry validity, instruction proxies, scenario-defined physical checks and offline agreement with released reference solids.

The contribution is an implemented workflow and empirical evaluation. Neither stronger general editing performance nor physical certification is established by the current experiment.

2. Related work

DeepCAD [7] studies parametric construction histories, Text2CAD [2] generates CAD from text, and CAD-Editor [8] studies instruction-based sequence editing. BenchCAD [9] releases programmatic mechanical tasks and edit pairs. We use public task and reference artifacts with a disclosed local split; results are not an official leaderboard submission or a reproduction of these systems.

Graph of Thoughts [1] motivates exploration of multiple candidate states, while UCT [3] balances value estimates and exploration in a tree. Our candidate values are imperfect verification proxies; classical search guarantees are not

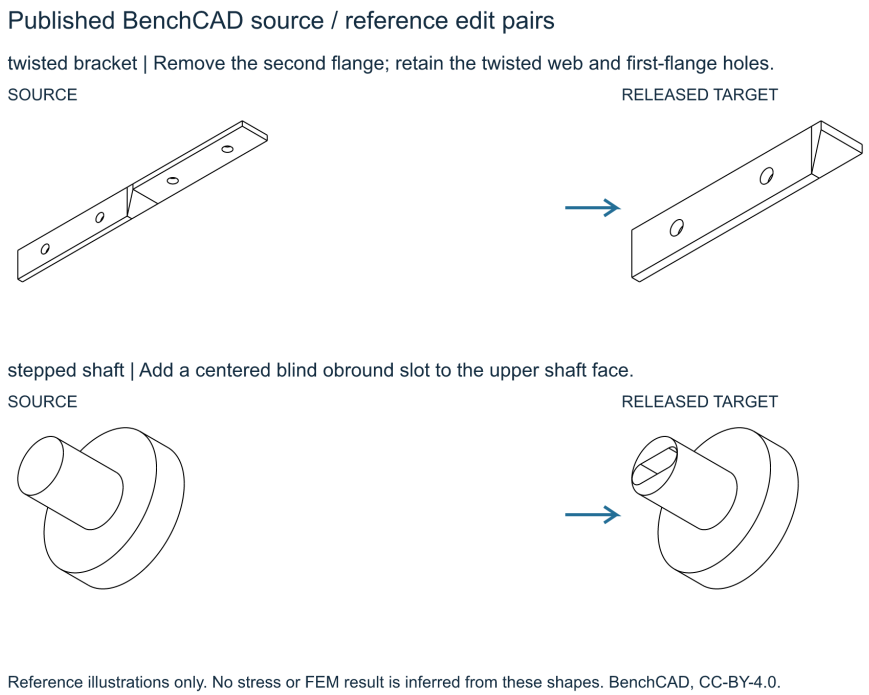


Figure 1. **Published mechanical CAD edit pairs held out in the local protocol.** Top: remove the perpendicular end flange while preserving the twisted web and the first-flange holes. Bottom: add a centered blind obround slot to the upper cylindrical shaft section. These are source/released-target illustrations, not predicted outputs. Both views are exported from solids. Candidate patches are evaluated separately with a fixed execution and reference-scoring protocol.

assumed for language-model proposals or these proxies.

FEM-Bench [4] and CADEngBench [6] connect generated engineering artifacts with numerical evaluation. Our finite-element checks operate under specified synthetic material, load and fixture contracts. They assess a declared scenario and do not recover the intended physical use of a public CAD edit task.

3. Astra-guided CAD patch repair

3.1. Original-source patch contract

Given source program C and edit instruction u, Astra proposes patch p, and an executor builds $\hat{C} = \mathcal{E}(C, p)$. Each operation contains a zero-based original line index, deletion count and replacement-line list. Operations must be typed, ordered, nonoverlapping and in bounds. CadQuery syntax or the native command schema is validated after application.

Repair revises the patch. Astra receives a failed candidate and a structured diagnosis, then generates a replacement patch whose indices still refer to C. Repairs are never silently concatenated onto an already modified program. The source and all candidate patches remain available in the trace; a verified selection exports executable CAD and a STEP solid.

3.2. Retrieval and graph exploration

Experience retrieval supplies prior CAD edits and, in the general implementation, successful repair trajectories. Task decomposition records requested changes and preserved features. Diverse hypotheses produce candidate CAD states. MCTS or best-first scheduling controls expansion, repair and backtracking within a fixed resource budget. Exact-state reuse must preserve model-visible context and edge statistics; similar volume or views alone are insufficient for merging.

Feasible candidates enter a bank. Pareto filtering compares declared objectives, followed by a fixed tie-break. Final robustness checks can return a candidate to search while budget remains. An empty verified bank yields an explicitly unverified fallback, not a certified solution. The recorded benchmark uses the narrower protocol in Section 5.2: its experience memory is frozen, and decomposition and repair-strategy retrieval are not exercised. Implementation capability and measured experiment scope are reported separately.

3.3. Hierarchical verification

The hierarchy comprises geometry, constraints, semantics, FEM and robustness. Geometry checks patch application, execution, positive volume and solid validity. Explicit

Source	Train	Validation	Test
BenchCAD	492	44	124
CAD-Editor	2,048	128	128
Total	2,540	172	252

Table 1. **Retained primary data.** Counts define our local protocol, not an official benchmark split. Source-specific reference construction remains distinct.

measurable constraints are checked on executed geometry; the recorded benchmark additionally uses numeric-literal heuristics from instructions. Semantic judgement uses Astra and rendered source/candidate views. These are imperfect signals, not access to a target program or proof of reference agreement.

Physical checks require declared units, material, fixture, load and acceptance criteria. The implementation records missing required contracts or inconclusive solves as unknown. The recorded benchmark uses a disclosed synthetic linear-elastic scenario, with its source-conditioned FEM gate specified in Appendix A. Robustness under selected perturbations is limited to those perturbations; passing does not certify manufacturing tolerance or operating safety.

3.4. Offline outcome measurement

Reference solids are reserved for final scoring. For predicted solid $\hat{B}$ and released or reconstructed target $B^\star$, volume intersection-over-union is

$$J_{\text{vol}} = \frac{V(\hat{B} \cap B^\star)}{V(\hat{B}) + V(B^\star) - V(\hat{B} \cap B^\star)}. \quad (1)$$

The local strict threshold is 0.99999. Patch validity, execution success and strict reference agreement are distinct outcomes. References never supply online search rewards or select a patch during inference.

4. Public data and local protocol

Two native representations. The retained public-data split contains 2,540 training, 172 validation and 252 test tasks (Table 1). BenchCAD provides CadQuery source/target programs, natural-language edits and released STEP references [9]. Microsoft's CAD-Editor provides native six-bit sketch–extrude edit sequences [8]. We retain their different representations and backends rather than treating them as interchangeable histories. CAD-Editor reference solids are reconstructed locally through our decoder; they are not author-released STEP references.

Provenance and isolation. BenchCAD is pinned at revision `5919f578ab09ec283603a082fab07c7639ab56eb` and declares CC-BY-4.0. Its 748 edit pairs yield 710 after family-based exclusion of pipe/electrical content. Family groups are joined when source/target AST templates match after numeric-literal masking; whole connected components are assigned to local splits before patch construction. The earlier retained test slice had 126 tasks; two additional pipe-related instructions are excluded from the primary experiment, yielding 124. CAD-Editor uses a documented subset of the public training/evaluation archive. Exact source/target connected components are isolated within each source; cross-source geometric ancestry is not established.

Representation and retention. The saved local split and its omissions are held fixed for the Astra comparisons. Comments and blank lines are normalized when preparing CadQuery patches. Original instructions are retained without adding target-line hints. Native command sequences use a separate schema and geometry decoder. The training partition supplies retrieval examples only; no weights are fitted by the active pipeline.

Evaluation boundaries. BenchCAD's edit release was originally held out; locally partitioning it into retrieval and evaluation subsets changes the benchmark contract. We report local held-out agreement, not official leaderboard or official-test generalization. The primary denominator includes all 252 retained tasks, including unavailable comparisons. Reference availability is tracked separately. A previously identified instruction/target conflict in a shaft-slot record illustrates that reference agreement does not certify natural-language compliance for every label. The graph-search analysis uses the 124 BenchCAD tasks, including one invalid reference in its overall denominator.

5. Experiments

5.1. Astra baseline and execution-feedback repair

We evaluate frozen GPT-6 Astra [5] on the retained public tasks with a fixed system prompt, original-coordinate patch contract and native geometry scorer. The saved single-answer baseline reaches 106/252 strict matches: 89/124 on BenchCAD and 17/128 on CAD-Editor. It uses medium reasoning effort and one sample per task. A separate execution-feedback repair round supplies validator or executor errors without references; it produces 107/252 strict matches. These results use the same retained task IDs and are separate from the graph experiment below.

5.2. Experience-guided graph search

Experience-guided graph search. Figure 2 shows the agentic pipeline we wrap around Astra, which remains

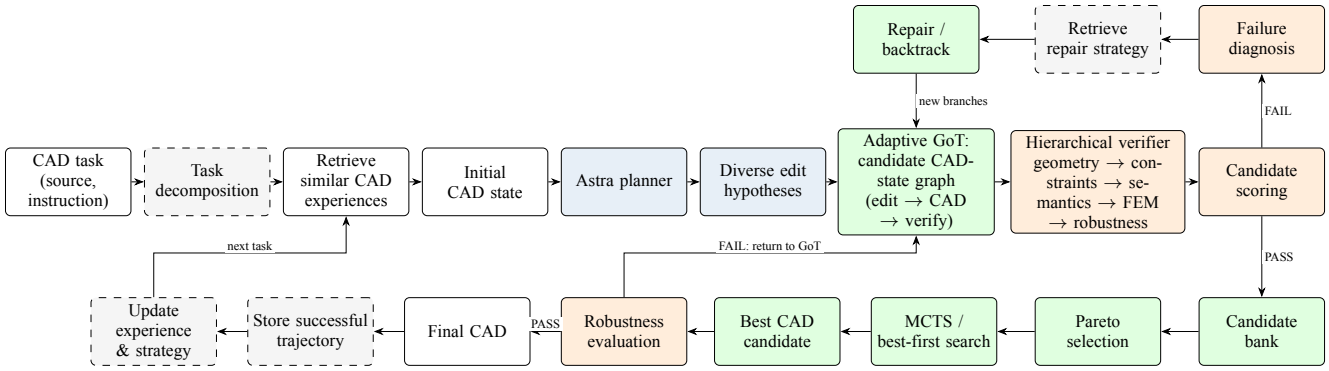


Figure 2. **Experience-guided graph search around a frozen Astra.** Astra (blue) plans and proposes edits; target-free checks (orange) score every candidate CAD state; search components (green) repair or backtrack failed states, keep passing ones in a candidate bank and select among them. No component is trained. Dashed stages are not exercised in the reported run: experience memory is frozen to BenchCAD training edits so that held-out tasks cannot inform one another, and the planner receives the full instruction without decomposition or retrieved repair strategies.

Configuration (BenchCAD, 124 tasks)	Strict	+	−	p
Astra, single answer	89	–	–	–
Retrieved experiences only†	94	7	2	0.18
GoT + geometry, constraint and FEM checks	89	1	1	1.00
+ MCTS repair and backtracking	88	1	2	1.00
+ Pareto selection	88	1	2	1.00
+ retrieved-experience candidate	88	1	2	1.00
+ robustness	88	1	2	1.00
+ semantics (full pipeline)	85	2	6	0.29

Table 2. **Pipeline components added in order** (Figure 2). Strict: volume IoU $\geq$ 0.99999 against released targets, scored offline. +/−: tasks gained/lost against the single Astra answer; p: exact two-sided McNemar test. †One low-effort Astra answer with two retrieved training edits as worked examples, before any search; identified after the results were known.

frozen. Retrieval supplies the two BenchCAD *training* edits most similar to the task (bag-of-words cosine over instruction and component name) as worked examples. The planner then proposes three alternative edit hypotheses in one low-effort call, each following a different reading where the instruction is ambiguous. With the original answer they form the nodes of a candidate CAD-state graph (3.3 distinct candidates per task on average). A hierarchical verifier checks every node, reading only the source program, instruction and candidates. *Geometry*: the patch applies and executes to a valid solid that differs from the source. *Constraints*: values requested in the instruction appear in the inserted lines. *Semantics*: when passing candidates disagree on geometry, Astra compares rendered isometric and top views of the source and each geometrically distinct candidate. *FEM*: a tetrahedral linear-elastic scenario check (fixed lower 5 % band, 1 N axial load on the upper band) must pass whenever the source part passes. *Robustness*: perturbing inserted numbers by ± 0.5 % must keep a valid solid with the same face count. A failing node receives a diagnosis naming the failed stage. MCTS with UCT ($c = 0.7$) then repairs the selected node using that diagnosis or, if it passes, backtracks and requests a hypothesis different from all passing ones. Search runs only on tasks with no passing candidate (19) or with geometrically disagreeing passing candidates (23), two expansions each; the $5 cap stopped it after 31 of these 42 tasks (29 repairs, 33 backtracks). Pareto selection keeps nodes passing the hard gates and chooses from the front over semantic vote, geometric consensus, instruction values and robustness, keeping the original answer on ties. References enter only the final scorer. Appendix A gives prompts and costs.

Search does not raise agreement. Table 2 adds components in pipeline order on the 124 BenchCAD held-out tasks; FEM is defined only for these programs, since CAD-Editor sequences carry no physical scale. Graph hypotheses with geometry, constraint and FEM checks keep strict agreement at 89/124, the same as the single Astra answer. Each later component leaves it at 88, until rendered-view semantics lowers it to 85 (2 gained, 6 lost, $p = 0.29$). No difference is significant. The whole pipeline costs $5.95 at list price.

Retrieval is the only component that helps, and only on its own. Astra's answer with two retrieved training edits, before any search, reaches 94/124 (7 gained, 2 lost, $p = 0.18$). We identified this configuration after seeing the results, so it is a hypothesis for a confirmatory run rather than an established gain. Inside the graph, the same retrieved answer competes with the other candidates, and selection rarely picks it.

Why verification cannot steer the search. The target-free checks hardly separate Astra's errors from its successes. Of its 35 incorrect BenchCAD answers, 24 pass the geometry, constraint and FEM stages, while 13 of its 89 correct answers fail at least one, mostly because a requested value is derived rather than written literally. An incorrect but valid edit that contains the requested values looks exactly like a correct one to these checks, so search has little signal to act on. Repair restores validity, not agreement. A deterministic repair layer re-indents inserted CadQuery lines and clamps, rounds and terminates CAD-Editor commands. It changes 19 of Astra's 252 answers, all CAD-Editor sequences, and makes 13 of them execute, but none reaches strict agreement.

6. Limitations and conclusion

ReSolve implements CAD editing as proposal, execution, diagnosis and patch repair around frozen Astra. The recorded graph pipeline reaches 85/124 strict matches against 89/124 for the original answer. Retrieval alone reaches 94/124, but its gain is post hoc and nonsignificant. These results support investigation of verifier quality rather than a claim of improved general editing performance.

The local split is not an official benchmark protocol. One recorded search run, unequal inference budgets, proxy constraints, a view-based semantic judge and scenario-specific FEM limit conclusions. Executable geometry may still implement the wrong feature; valid physical response under a synthetic load does not establish the intended use. Future comparisons should fix candidate-generation and verification budgets, use independently defined feature checks and report multiple seeds. The implementation supports broader decomposition and memory-update stages, but unexercised capabilities are not attributed to the reported experiment.

References

[1] Maciej Besta, Nils Blach, Ales Kubicek, et al. Graph of thoughts: Solving elaborate problems with large language models, 2023. 1

[2] Mohammad Sadil Khan et al. Text2CAD: Generating sequential CAD designs from beginner-to-expert level text prompts. In *Advances in Neural Information Processing Systems*, 2024. 1

[3] Levente Kocsis and Csaba Szepesvári. Bandit based monte-carlo planning. In *Machine Learning: ECML 2006*, pages 282–293. Springer, 2006. 1

[4] Saeed Mohammadzadeh, Erfan Hamdi, Joel Shor, and Emma Lejeune. Fem-bench: A structured scientific reasoning benchmark for evaluating code-generating llms, 2025. 2

[5] OpenAI. GPT-6 Astra model documentation, 2026. Model `gpt-6-astra`, accessed 4 October 2026. 3

[6] Harmanjot Singh, Abhra Dubey, and Jorge Alejandro Amador Herrera. Cadengbench: It looks like cad, but does it work? evaluating parametric design, assembly reasoning, and physics simulation, 2026. 2

[7] Rundi Wu, Chang Xiao, and Changxi Zheng. DeepCAD: A deep generative network for computer-aided design models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021. 1

[8] Yu Yuan, Shizhao Sun, Qi Liu, and Jiang Bian. Cad-editor: A locate-then-infill framework with automated training data synthesis for text-based cad editing, 2025. 1, 3

[9] Haozhe Zhang, Kaichen Liu, Miaomiao Chen, Lei Li, Shaojie Yang, Cheng Peng, and Hanjie Chen. Benchcad: A comprehensive, industry-standard benchmark for programmatic cad, 2026. 1, 3

A. Experience-guided graph search protocol

This section specifies the run behind Figure 2 and Table 2, on the 124 BenchCAD held-out tasks. Every new call uses `gpt-6-astra` at low reasoning effort with a 2,500-token completion cap (2,000 for semantic judgement). The initial node of each task is the saved medium-effort answer of Section 5.2. A hard budget at list price ($10 and $50 per million input and output tokens) refuses any call whose worst-case cost would exceed its cap: $5 for hypotheses and search, and $4 more for retrieval and semantics. Responses are saved and never re-billed. The calls cost $3.08 for hypotheses (124 calls), $1.14 for search (62), $1.19 for retrieval (124) and $0.53 for semantics (21), $5.95 in total.

Experience retrieval. The memory holds the 492 retained BenchCAD training edits and is never updated during evaluation. Each task and memory item is represented by lowercase word counts of its instruction and component name. The two nearest items by cosine similarity are shown as worked examples, each with its instruction, the changed original lines with zero-based indices and its correct patch, followed by the task input under the system prompt of the frontier comparison.

Hypotheses. One call per task returns three alternative edits under the following system prompt:

```
You edit mechanical CAD programs. Propose exactly 3
    alternative edits for the instruction. Where the
    instruction is ambiguous (which feature, edge set,
    reference dimension or construction is meant), make
     each alternative follow a different reading; if it
     is unambiguous, vary the implementation. Return
    only JSON {"candidates":[{"interpretation":"one
    sentence","edits":[{"start":0,"delete":0,"insert":[
    "replacement line"]}]}, ...]}. Indices are zero-
    based lines of the ORIGINAL input; operations must
    be ordered, nonoverlapping and in bounds. Preserve
    unrelated features. Return no other text.
```

Verifier stages. *Geometry* parses the answer, applies the patch under the edit contract and executes the program in an isolated CadQuery worker; the result must be a valid solid whose volume IoU with the source is below 0.99999. *Constraints* extract numbers from the instruction and require each to appear in the inserted lines, accepting "from a to b", "by x" and radius/diameter forms. Candidates are also clustered by executed geometry (IoU ≥ 0.99999); cluster size is the consensus. *FEM* meshes the exported STEP solid with gmsh at element size $\max(d/18, 0.5)$ mm for bounding-box diagonal d. It solves linear tetrahedral elasticity ($E = 210$ GPa, $\nu = 0.3$), with the lowest 5 % of the height fixed and a 1 N axial load on the top 5 %. It requires a single anchored component, positive compliance and force, moment and energy balance within 10^{-6}. A candidate must pass when the source part passes. *Robustness* scales every number in the changed lines by 1 ± 0.005 and reports the fraction of perturbed programs that still give a valid solid with the source candidate's face count. *Semantics* applies only where passing candidates form several geometry clusters. One matplotlib render per cluster shows the source beside the candidate in isometric and top views. Astra returns the best-matching candidate as JSON.

Search. A node's value is 0 if geometry fails, and otherwise $0.4 + 0.2\,[\text{FEM}] + 0.2\,[\text{values}] + 0.2\,[\text{changed}]$. UCT selects the node maximizing $\bar{v} + 0.7\sqrt{\ln N/n}$, with N the total and n the node's visit count. If the selected node's mean value is below 1, the expansion is a repair. The node's answer is returned with its diagnosis, which names the failed patch application or execution, the FEM failure, missing requested values or an unchanged solid. Otherwise the expansion backtracks: it lists all passing alternatives and requests one meaningfully different edit. The new node is evaluated and its value backed up along the path to the root. Tasks with no passing candidate are searched first, then tasks whose passing candidates disagree on geometry; tasks with unanimous passing candidates are not searched.

Selection. Hard gates are geometry, FEM (when the source passes), no contradicted instruction value and a changed solid. Among passing nodes, the Pareto front is taken over semantic vote, consensus, presence of instruction values and robustness. The node with the lexicographically largest objective vector is chosen, and ties keep the original answer. If no node passes, the original answer is kept.

Deterministic repair. For the 252 saved Astra answers we also run a monotone repair loop. A diagnosis records the furthest stage reached (parse, apply, execute, valid solid). Candidate fixes either re-indent or de-indent inserted CadQuery lines, or, for CAD-Editor commands, clamp coordinates to the six-bit range, clamp rotation entries to $[-1, 1]$, round non-integers, trim surplus curve fields and insert missing terminators. A fix is accepted only if it reaches a later stage, for up to three rounds.